# DoT Front-end

Design and progress

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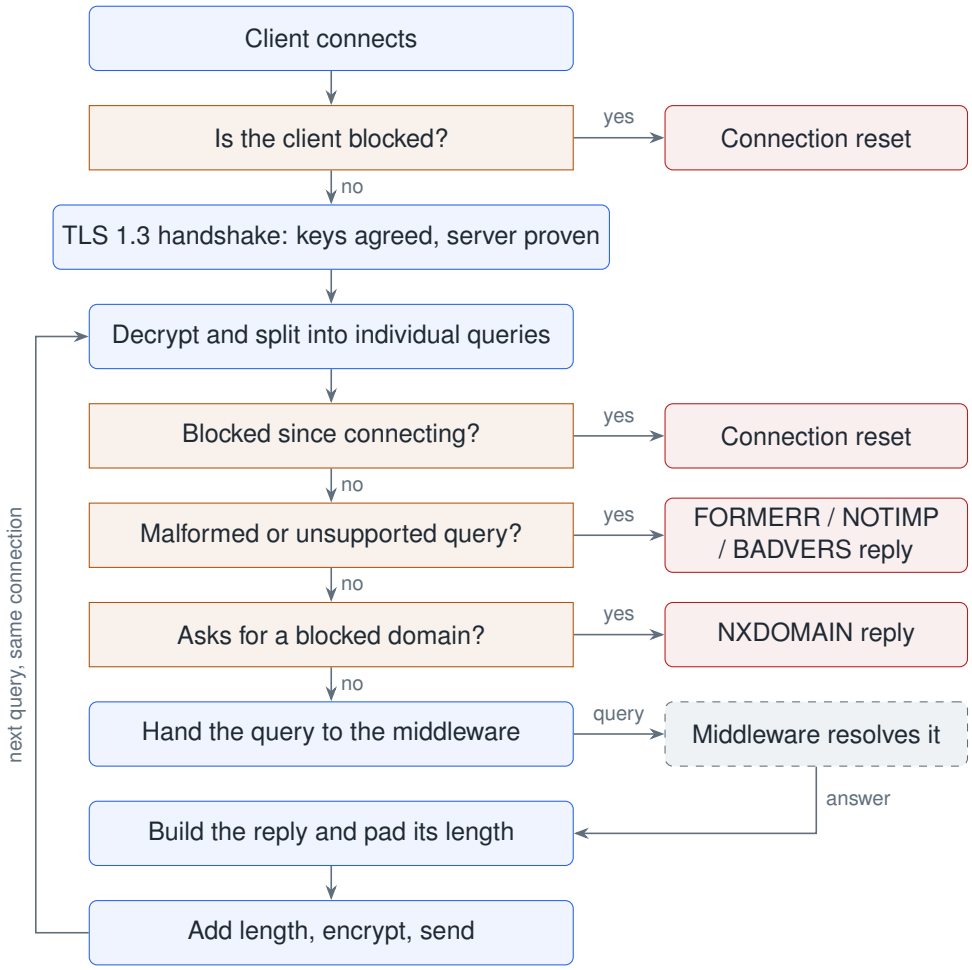
The front-end is the resolver's front door. It accepts encrypted DNS queries from clients over DNS-over-TLS (TCP port 853), checks each one, passes valid queries to the middleware, and sends back the encrypted answers. It also decides which clients and queries get rejected, and how.

**All 11 features are complete and tested**, including end to end with `dig +tls`.

## How a query flows through the front-end




A connection stays open after a reply, so one client can send many queries without repeating the TLS handshake. It may send several without waiting (pipelining); each answer goes out as soon as it is ready, matched to its query by ID.

## Features

| Feature | What it does | Status |
|---|---|---|
| **Query parsing and validation** | Turns raw bytes into a checked query; rejects anything malformed or hostile without ever reading outside the message | **Done** |
| **Name checks** | Accepts names made of letters, digits, - and _; compares them without regard to case, but echoes the client's own spelling back | **Done** |
| **Reply building** | Turns the middleware's answer (answer, authority and additional records) into a correct DNS reply, always producing a valid message (falls back to SERVFAIL if needed) | **Done** |
| **Length padding** | Pads replies to a fixed block size (468 bytes) so encrypted traffic doesn't reveal which site was looked up by its size | **Done** |
| **Message framing** | Splits the decrypted byte stream into individual queries, and prefixes each reply with its length | **Done** |
| **TLS encryption** | Encrypts every connection with OpenSSL, TLS 1.3 only; proves the server's identity with its certificate | **Done** |
| **Abuse blocking** | Counts misbehaviour per client; a client that misbehaves too often is cut off before any TLS work, for a set time | **Done** |
| **Policy rejections** | Answers NXDOMAIN for queries about blocked domains, read from a file | **Done** |
| **Connection handling** | Serves many clients at once, several queries per connection, with timeouts and limits | **Done** |
| **Configuration** | Listen address, certificate and key, limits, blocking settings, blocked-domain list | **Done** |
| **Runnable server** | `dns_server <config-file>`, tested end to end with `dig +tls` | **Done** |

## How the front-end rejects queries

| Situation | Response | Why |
|---|---|---|
| Client is blocked for abuse | Connection reset, no reply, no TLS work | No work or bandwidth spent on attackers |
| Message too broken to read its ID, or not a query at all | Connection closed, no reply | There is nothing valid to reply to |
| Malformed query | FORMERR | Tells a buggy client what's wrong |
| Query kind we don't support | NOTIMP | A clear answer instead of silence |
| EDNS version we don't support | BADVERS | The client retries with version 0, keeping EDNS and padding |
| Query asks for a blocked domain | NXDOMAIN, connection stays open | The name looks nonexistent, so the client doesn't try another resolver; it isn't punished |
| Middleware overloaded, or client over its rate limit | SERVFAIL / REFUSED | The client can retry later |

## What the response codes mean

Every DNS reply carries a response code. FORMERR, NOTIMP, BADVERS and REFUSED say the problem is with the query; SERVFAIL says the problem is on the resolver's side, so retrying later can succeed.

| Code | What it means in this resolver | Retry? |
|---|---|---|
| **FORMERR**<br>Format error · code 1 | **The query is broken.** It was readable enough to reply to, but breaks DNS rules: wrong section counts, a damaged name or one with characters outside letters, digits, − and _, a bad EDNS record, or extra bytes at the end | No |
| **NOTIMP**<br>Not implemented · code 4 | **Valid, but unsupported.** An operation other than a standard query, a class other than Internet (e.g. CHAOS), or a record type outside A, NS, CNAME, SOA, PTR, MX, TXT, AAAA and ANY (e.g. HTTPS) | No |
| **BADVERS**<br>Bad EDNS version · code 16 | **Wrong EDNS version.** The client used a version above 0, the only one that exists; the reply says "use version 0" | Yes, with v0 |
| **NXDOMAIN**<br>Name error · code 3 | **The name does not exist.** The upstream says so, or the domain is on the blocked list | No |
| **REFUSED**<br>Refused · code 5 | **Understood, but not allowed.** The client is over the middleware's rate limit, or blocked | No |
| **SERVFAIL**<br>Server failure · code 2 | **No answer could be produced.** The query was fine, but the middleware was overloaded, the upstream servers failed, or the answer couldn't be turned into a valid reply | Yes, later |

## Abuse blocking

Each client earns **strikes** for misbehaving. Too many strikes within a time window and its address is blocked: every new connection from it is reset before the TLS handshake, and an open connection is reset at its next query. Blocks expire on their own.

| Counts as a strike | Doesn't count |
|---|---|
| Being refused by the rate limiter (*rate-limit strike*) | Asking for a blocked domain |
| A failed TLS handshake: not TLS, or a version we refuse (*protocol strike*) | An unsupported query (NOTIMP) or EDNS version (BADVERS) |
| A zero-length message, one with no readable ID, or a reply instead of a query (*protocol strike*) | Going quiet until the idle timeout |
| A malformed query, FORMERR (*protocol strike*) | Hanging up mid-handshake, or rejecting our certificate with an alert |

Protocol strikes have their own, **higher** limit, because honest clients occasionally fail a handshake or send something odd. IPv6 clients are counted per /64 network, since one device usually owns a whole /64. A client reaching us as `::ffff:1.2.3.4` is the same client as `1.2.3.4`.

## Limits that protect the server

| Limit | Default | What happens |
|---|---|---|
| Connections at once | 1000 | Extra connections are closed straight away |
| Connections from one source | 32 | Extra connections are closed straight away. IPv4 per address, IPv6 per /64 |
| Idle connection | 10 s | Closed; also covers a client that never finishes its handshake or a message. Only whole queries count as activity, not stray bytes. A connection still waiting for an answer is never idle |
| Answers left unread | 10 s | A client that reads none of its waiting answers for the idle timeout is reset |
| Queries waiting per connection | 100 | The server stops reading from that client until answers go out |
| Unsent answers per connection | 256 KiB | A client that stops reading its answers is closed |
| Clients tracked for strikes | 100,000 | When full, new clients aren't tracked; existing blocks always stay |

## How the encryption works

- **TLS 1.3 only.** Every TLS 1.3 cipher suite uses an ephemeral key exchange and authenticated encryption; older versions are refused.
- **The session key is never sent.** Client and server each compute it from a key exchange (ECDHE), so someone watching the network can't obtain it.
- **The server proves its identity** by signing the handshake with its private key. The key never leaves the server, and the signature stops anyone impersonating it.
- **Forward secrecy.** A stolen private key would let an attacker impersonate the server in future, but not decrypt traffic recorded earlier.
- **Length padding** covers the one thing TLS doesn't hide: message sizes.
- **Key file protection.** The server warns at startup if the private key can be read by other users, and `*.pem` / `*.key` files are kept out of git.
- **ALPN.** A client offering the `dot` protocol name gets it confirmed; it isn't required.

## What the front-end uses from the middleware

The middleware is built by a teammate; the front-end calls three of its functions:

- `submit(name, type, client, callback)` hands over a query. The callback runs exactly once with the answer, possibly on another thread; the front-end passes the encoded reply back to its own I/O thread.
- `is_blocked(address)` answers "is this client blocked right now?", at connect time and before each query.
- `report_abuse(address)` records a protocol strike.

## Configuration

These keys go in the same config file as the middleware's (see `config/middleware.example.conf`):

| Key | Default | Meaning |
|---|---|---|
| `listen` | `0.0.0.0:853` | Address and port; `[::]:853` also accepts IPv4 |
| `tls_cert`, `tls_key` | none | Certificate chain and private key (PEM). Both are required to run the server |
| `max_connections` | 1000 | Connections at once |
| `max_connections_per_source` | 32 | Connections from one IPv4 address or IPv6 /64; 0 means no limit |
| `idle_timeout_ms` | 10000 | Idle time before a connection is closed, and how long answers may sit unread |
| `blocked_domains_file` | none | One domain per line, `#` comments allowed. A domain also blocks everything under it, so a lone `.` refuses every query |
| `block_rate_limit_strikes` | 0 (off) | Rate-limit strikes before a block |
| `block_protocol_strikes` | 0 (off) | Protocol strikes before a block; must be higher than the rate-limit one |
| `block_window_ms` | 10000 | How long a strike is remembered |
| `block_duration_ms` | 300000 | How long a block lasts |
| `block_max_entries` | 100000 | Clients tracked for strikes |

## Building, running and testing

Needs Linux, `g++-14`, CMake and OpenSSL 3 (`sudo apt install g++-14 libssl-dev`). On other systems everything except the DoT server itself still builds.

```
cmake --preset local && cmake --build build/local    # build
ctest --test-dir build/local                         # run all tests

# a self-signed certificate for local testing
openssl req -x509 -newkey ec -pkeyopt ec_paramgen_curve:P-256 -nodes \
        -keyout server.key -out server.pem -days 30 -subj /CN=localhost
chmod 600 server.key

build/local/dns_server my.conf                       # with tls_cert/tls_key set
dig @127.0.0.1 -p 853 +tls example.com               # query it
```